

Requirement Analysis

Technology Stack (Architecture & Stack)

Date	24 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Technical Architecture

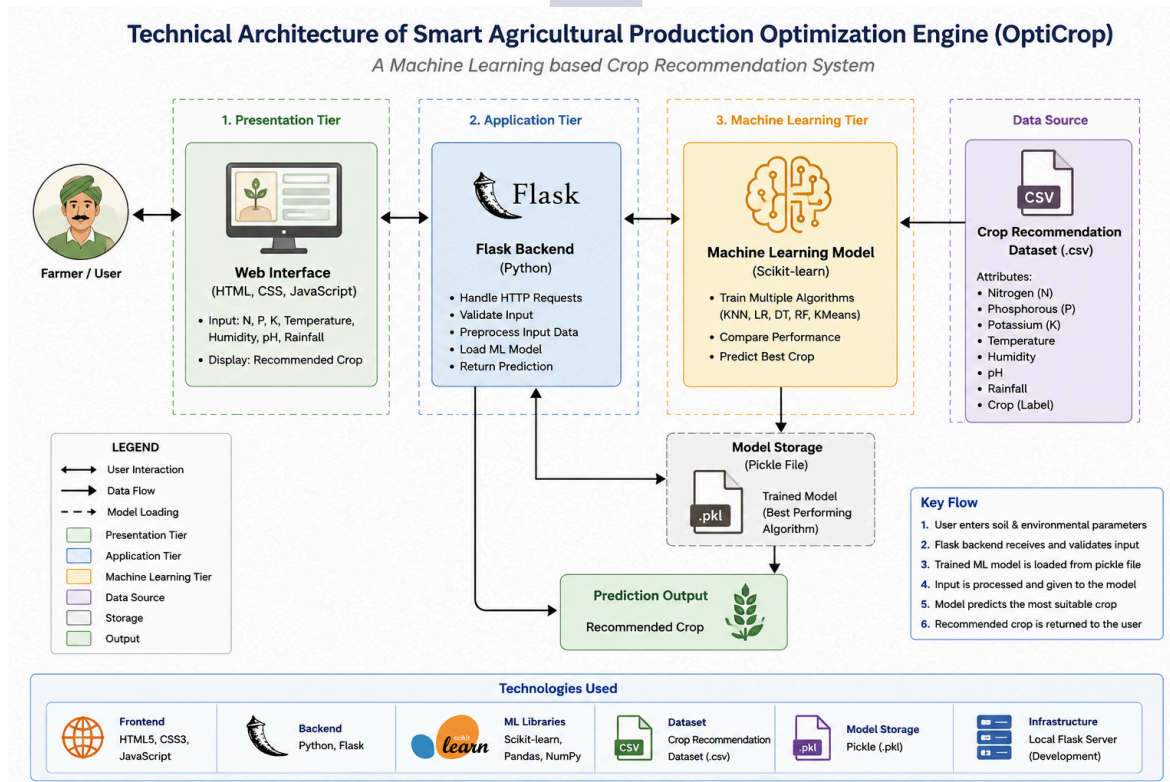


Figure 1: Technical Architecture of Smart Agricultural Production Optimization Engine

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for entering soil and environmental parameters and displaying crop recommendations	HTML5, CSS3, JavaScript
2	Application Logic	Handles HTTP requests, input validation, and communication between frontend and machine learning model	Python, Flask
3	Data Processing	Loads dataset, preprocesses data, performs train-test split, and prepares features for model training	Pandas, NumPy, Scikit-learn
4	Machine Learning Model	Trains multiple ML algorithms, compares their performance, and predicts the most suitable crop	Scikit-learn (Random Forest, Decision Tree, KNN, Logistic Regression, K-Means)
5	Dataset	Stores agricultural training data containing soil and environmental parameters	CSV Dataset (Crop Recommendation Dataset)
6	Model Storage	Stores the trained machine learning model for future predictions	Pickle (.pkl)
7	File Storage	Stores dataset, trained model, templates, and static resources	Local File System
8	External API	Not required for the current implementation	N/A
9	Database	No database is required for this project as predictions are	N/A

		generated directly from the trained model	
10	Infrastructure	Hosts the Flask application and machine learning model	Local Python Server (Flask Development Server)

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Frameworks and libraries used to develop the application	Flask, Scikit-learn, Pandas, NumPy
2	Security Implementation	Basic input validation to prevent invalid values from being processed	Flask Input Validation
3	Scalable Architecture	Modular architecture separating presentation, application logic, and machine learning components	Three-Tier Architecture
4	Availability	Application can run locally with continuous availability while the server is active	Flask Local Server
5	Performance	Pre-trained model loaded from a Pickle file eliminates retraining and provides fast predictions	Scikit-learn, Pickle